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Professional Summary

Senior Application Developer with 15+ years of experience in enterprise application development, system modernization, and backend engineering. Currently transitioning into Machine Learning with hands-on experience in building predictive models, handling imbalanced datasets, and model evaluation. Strong foundation in data processing, scalable systems, and problem-solving.

Technical Skills

Machine Learning: Supervised Learning, XGBoost, Random Forest, Decision Trees, Logistic Regression, Model Evaluation (Precision, Recall, F1-score, ROC-AUC), Feature Engineering, SMOTE, SHAP

Programming & Data Tools: Python, Pandas, NumPy, Scikit-learn, Java, SQL

Frameworks & Technologies: Spring Boot, REST APIs, Microservices, Apache Kafka, Apache Ignite, Apache Camel

Tools: GitHub, Jenkins, JIRA, Splunk, Dynatrace, Maven, JUnit, Mockito

Machine Learning Projects

Fraud Detection using Machine Learning

- Developed a fraud detection model using XGBoost to classify fraudulent transactions
- Handled imbalanced dataset using SMOTE to improve fraud detection capability
- Performed feature engineering and data preprocessing to enhance model performance
- Tuned hyperparameters using GridSearchCV for optimized results
- Evaluated model using Precision, Recall, F1-score, and Confusion Matrix
- Applied SHAP for model explainability and feature importance analysis
- Focused on improving recall to reduce false negatives in fraud detection

Titanic Survival Prediction

- Built an end-to-end machine learning application to predict Titanic passenger survival using XGBoost with a full preprocessing pipeline (imputation, scaling, encoding)
- Integrated SHAP (SHapley Additive Explanations) to provide feature-level interpretability, including waterfall plots to explain individual predictions

- Developed and deployed a FastAPI backend with /predict and /shap endpoints, enabling real-time inference and explainability via API
- Created two Streamlit applications: one using direct model inference and another using API-based architecture, demonstrating production-style system design
- Deployed the application using Render (FastAPI) and Streamlit Cloud, handling real-world issues like API latency, data transformation, and JSON-to-numpy conversions
- Built predictive models using Decision Tree and Random Forest algorithms
- Performed data cleaning, missing value handling, and feature engineering
- Created features such as IsAlone to improve prediction accuracy
- Evaluated models using classification metrics and confusion matrix
- Improved performance through hyperparameter tuning

Professional Experience

Application Developer / Technology Lead (Client Project)

Atos Syntel / Infosys | Oct 2019 – Present

- Collaborated with product owners and cross-functional teams in backlog grooming and sprint planning
- Led modernization of legacy mainframe systems (COBOL, DB2, JCL) into microservices-based Java architecture
- Designed and developed REST APIs using Spring Boot
- Built high-performance applications using Apache Ignite (in-memory computing)
- Developed and enhanced application features to resolve production issues
- Performed root cause analysis using Splunk and Dynatrace to reduce recurring defects
- Implemented unit and integration testing using JUnit and Mockito
- Collaborated with DevOps teams for CI/CD integration and production deployment
- Managed source code using GitHub with version control best practices
- Delivered demos to stakeholders and participated in sprint retrospectives

Data / ML Exposure:

- Developed Python scripts for data analysis supporting finance and data teams
- Applied data preprocessing techniques using Pandas and NumPy
- Built machine learning models as part of upskilling initiatives

Senior Associate

Cognizant | Feb 2009 – Oct 2019

- Developed scalable REST APIs and microservices using Java and Spring Boot
- Designed and maintained automation frameworks using Selenium, Appium, and Cucumber (BDD)

- Automated web and mobile applications (iOS and Android), improving testing efficiency
- Built reusable automation frameworks using Java and OOP principles
- Analyzed production logs using Splunk and resolved application defects
- Integrated automation frameworks with CI/CD pipelines using Jenkins and Docker
- Executed cross-browser testing using BrowserStack and Sauce Labs
- Developed SQL queries for backend data validation and analysis
- Worked on mainframe systems (COBOL, JCL, DB2, VSAM, CICS) for backend validation
- Collaborated with QA, DevOps, and business teams for successful releases
- Supported enterprise clients across banking and retail domains

Education

Bachelor of Engineering in Electronics and Communication

Los Angeles is one of the largest cities in the United States and is located in the state of California. It is famous for the entertainment industry and is home to Hollywood. Many movie studios and celebrities live and work in this area. The city enjoys warm weather for most of the year and attracts millions of tourists annually. Popular attractions include Universal Studios Hollywood, Griffith Observatory, and the Hollywood Walk of Fame. Los Angeles has beautiful beaches, diverse neighborhoods, and a strong international culture.

San Diego is another major city in California located near the border with Mexico. It is known for its pleasant climate, beautiful coastline, and relaxed lifestyle. The city is home to Balboa Park and the world-famous San Diego Zoo. Visitors enjoy beaches, outdoor activities, and water sports throughout the year. The local economy is supported by tourism, biotechnology, healthcare, and military operations.

San Francisco is located in northern California and is famous for its hills, cable cars, and the iconic Golden Gate Bridge. The city played a major role in the growth of the technology industry due to its proximity to Silicon Valley. Popular attractions include Alcatraz Island and Fisherman's Wharf. The city is known for innovation, diversity, and a vibrant cultural scene.

Dallas is a large metropolitan city in Texas. It is a major business, financial, and transportation center in the southern United States. The city has a strong economy supported by technology, healthcare, telecommunications, and banking industries. Dallas is also known for professional sports teams, shopping districts, and modern infrastructure. Visitors often explore museums, entertainment venues, and historical sites throughout the city.

Hartford is the capital city of Connecticut. It is one of the oldest cities in the United States and has a long history in the insurance industry. Many major insurance companies have headquarters or significant operations in Hartford. The city contains historical

landmarks, cultural institutions, and educational centers. Hartford offers a blend of New England tradition and modern business activity.

Austin is the capital of Texas and is widely known for technology, music, and innovation. Major technology companies have established offices in the city. Austin hosts many festivals, including the famous South by Southwest. The city has experienced rapid population growth over recent decades. Residents enjoy parks, lakes, outdoor recreation, and a thriving food culture.

New York City is the largest city in the United States and is located in New York. It is considered one of the world's most important financial, cultural, and media centers. Famous landmarks include Statue of Liberty, Times Square, and Central Park. The city consists of five boroughs and hosts people from nearly every country. Wall Street, Broadway, museums, and skyscrapers contribute to its global reputation.

New Jersey is a state located between New York and Pennsylvania. Many residents commute to New York City for work. The state offers beaches, suburban communities, parks, and industrial centers. The Jersey Shore attracts visitors during summer months. New Jersey has strong pharmaceutical, logistics, and financial industries.

McLean is a community in Virginia located near Washington, D.C. It is known for its affluent neighborhoods and proximity to federal government agencies. Many executives, diplomats, and government employees live in the area. Several major corporations maintain offices nearby. McLean provides convenient access to both urban amenities and suburban living.

Washington, D.C. is the capital of the United States. It serves as the center of the federal government. Major landmarks include the White House, United States Capitol, and Lincoln Memorial. The city contains numerous museums operated by the Smithsonian Institution. Millions of tourists visit annually to learn about American history and government.

Seattle is located in the state of Washington. It is known for technology, aerospace, and coffee culture. Major companies such as [Amazon](#) and [Microsoft](#) have strong connections to the region. Landmarks include the Space Needle and Pike Place Market. The city is surrounded by mountains, forests, and water, making it attractive for outdoor enthusiasts.

Denver is the capital of Colorado. It is located near the base of the Rocky Mountains. Denver is known for outdoor recreation, a growing technology sector, and a high quality of life. Residents enjoy skiing, hiking, and mountain activities throughout the year. The city combines urban development with easy access to nature.

Rocky Point, also known as Puerto Peñasco, is a coastal resort town in northwestern Mexico. It is located on the Gulf of California and is popular among visitors from Arizona and nearby U.S. states. The area offers beaches, fishing, boating, and water

sports. Tourism is a major source of income. Many travelers visit Rocky Point for weekend vacations and family getaways.

Pittsburgh is a city in Pennsylvania known for its industrial history and transformation into a center for healthcare, education, and technology. The city is home to respected universities and research institutions. Numerous bridges cross its rivers, creating a distinctive skyline. Pittsburgh has a strong sports culture and a growing technology ecosystem.

Cleveland is located in the state of Ohio along the shores of Lake Erie. The city has significant healthcare, manufacturing, and cultural institutions. The Rock and Roll Hall of Fame is one of its most famous attractions. Cleveland also offers parks, museums, theaters, and professional sports teams.

Niagara Falls is one of the most famous natural attractions in North America. It is located on the border between the United States and Canada. Millions of visitors come every year to witness the powerful waterfalls. Boat tours allow visitors to experience the falls up close. The surrounding region includes parks, hotels, restaurants, and entertainment venues. Niagara Falls is both a major tourist destination and an important source of hydroelectric power. The natural beauty of the falls makes it one of the most photographed landmarks in the world.